

252-0027

Introduction to Programming Exercises

Week 2: Git and EBNF

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Organization

- My name: *Jan Leonard*
- For questions: leonardj@student.ethz.ch
 - Please put «[EProg26]» in the subject
- New exercises: **Tuesday evening** (normally)
- Submit exercises by **Tuesday evening (23:59)** following week
 - Always submit via Git
 - Solutions in a separate Git project



Eprog TA — HS 2026

janleonard115.github.io

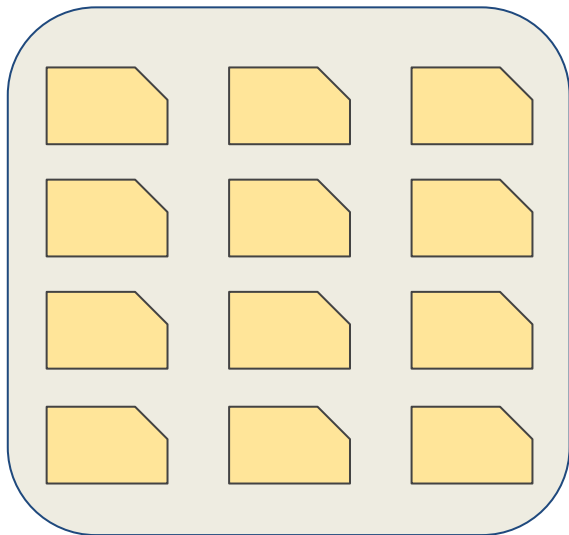
Plan For Today

- Live Demo of Git and GitHub
- Submission Demo
- EBNF Theory
- EBNF Extra Exercises
- Kahoot
- Disclaimer

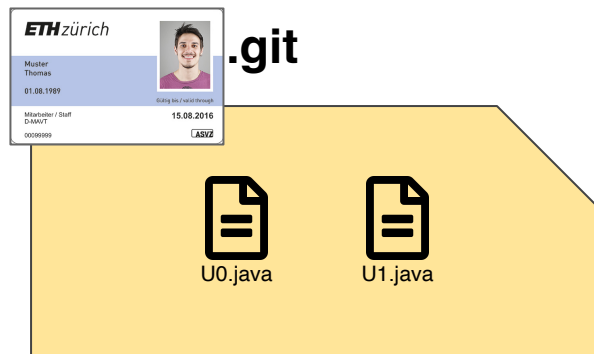
LIVE GIT AND GITHUB DEMO

LIVE SUBMISSION DEMO

Git Repository

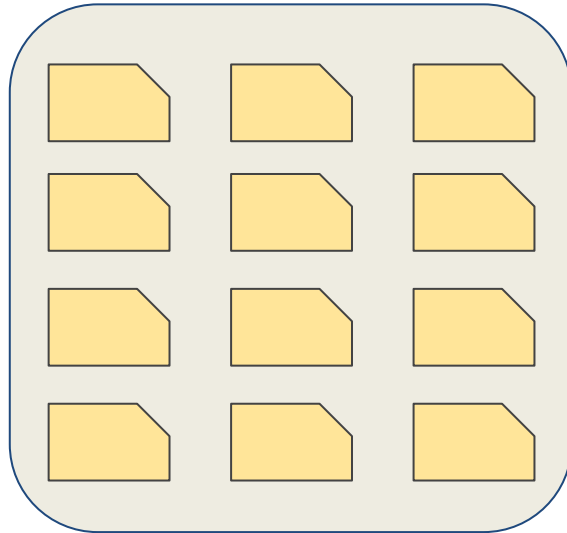


ETH Git-Server

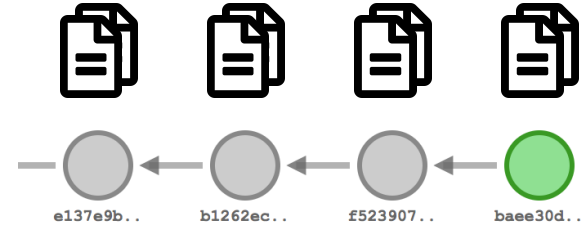


Every repository on the Git server is private

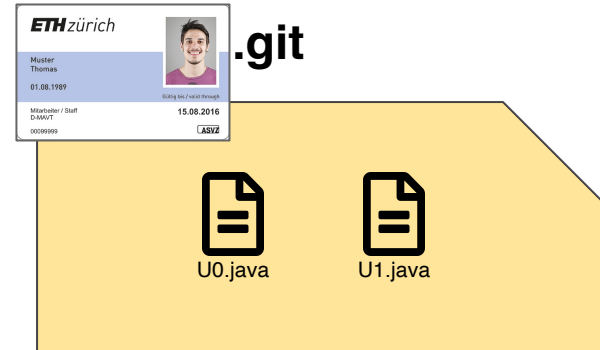
Git Repository



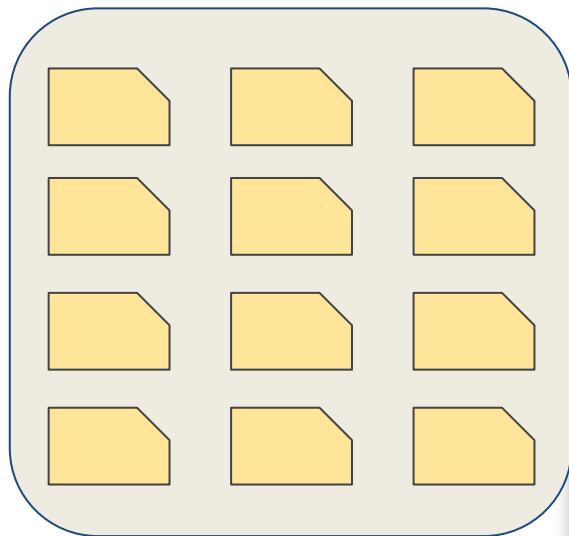
ETH Git-Server



Every repository on the Git server
Contains a sequence of commits (the history)



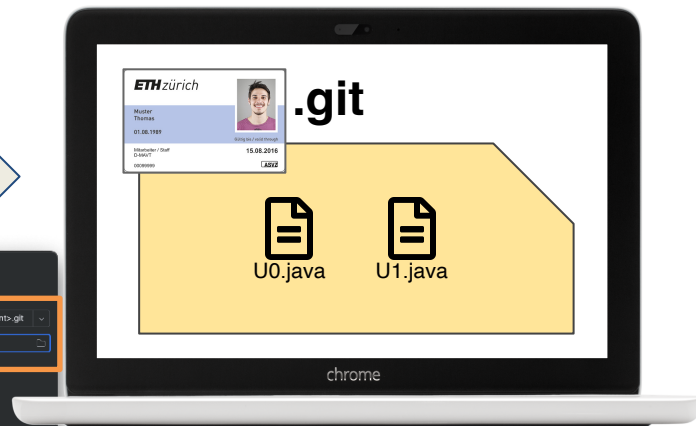
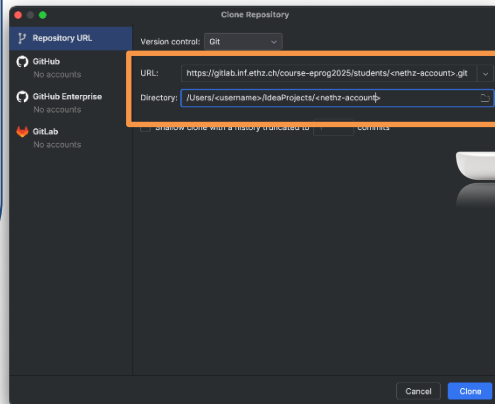
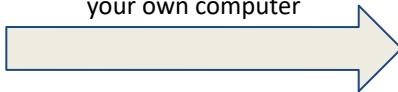
Git Clone: One-time setup



ETH Git-Server

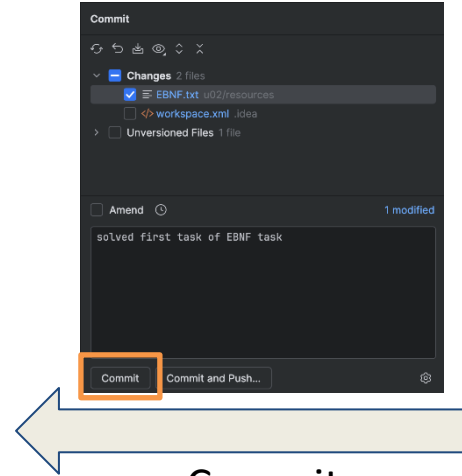
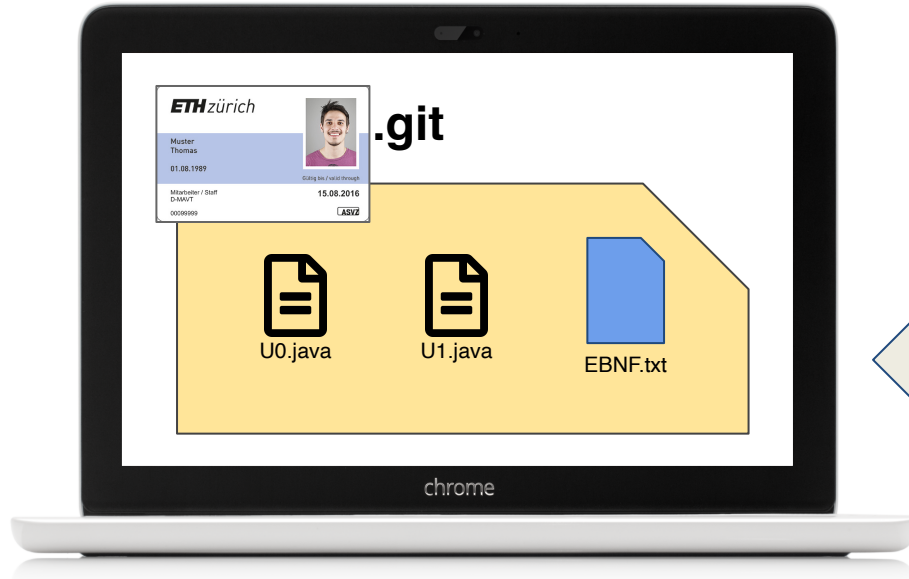
Clone

Copies the whole repository to
your own computer



Local Git repository

Git Commit: Save progress



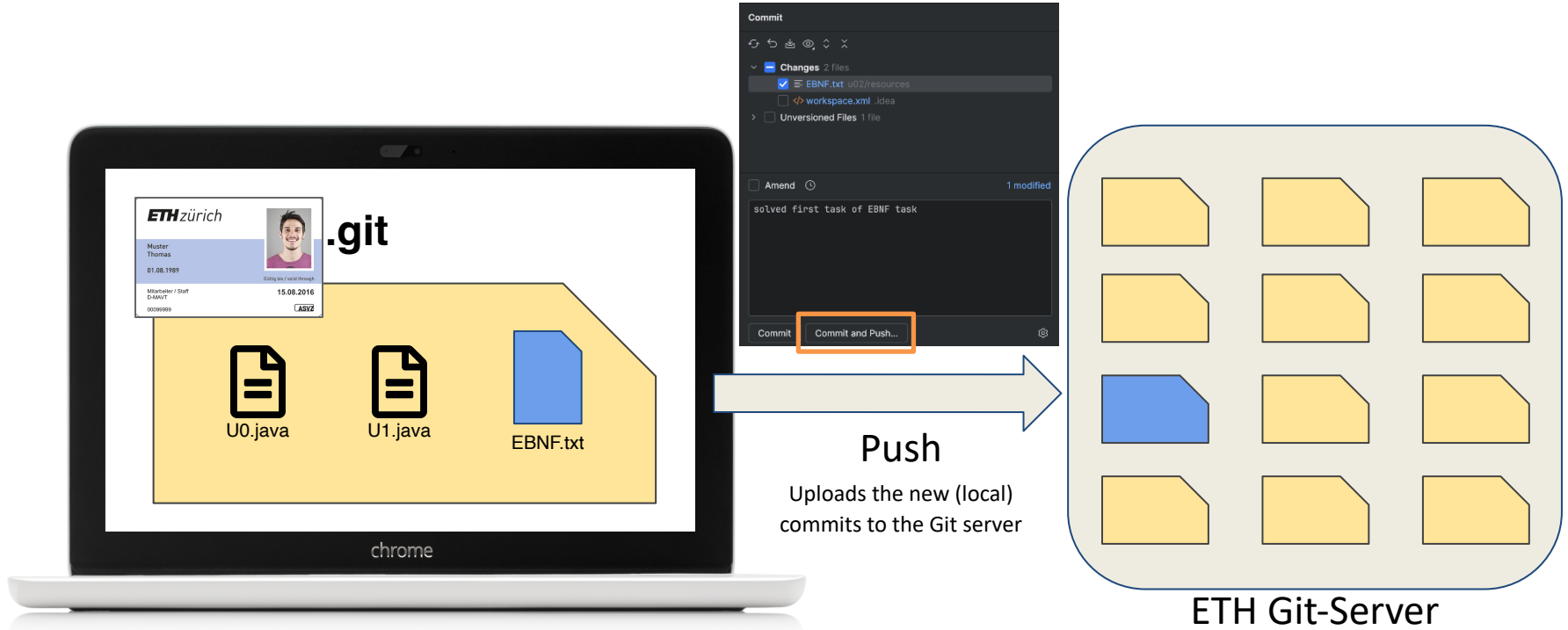
Commit

Adds a new commit with changes/new files to the local history



EBNF.txt

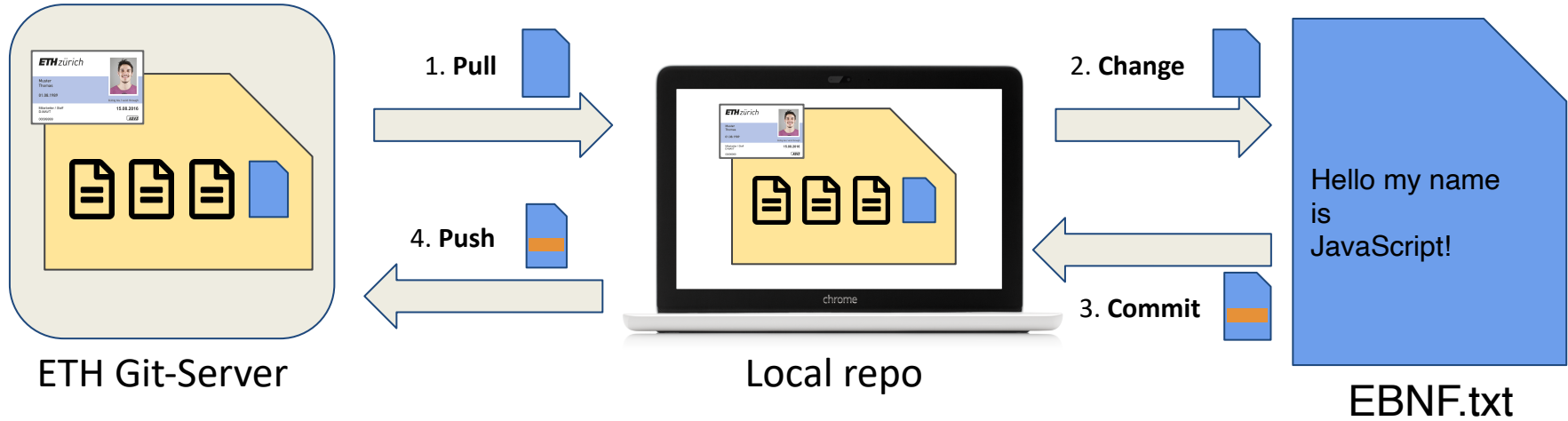
Git Push: Submit



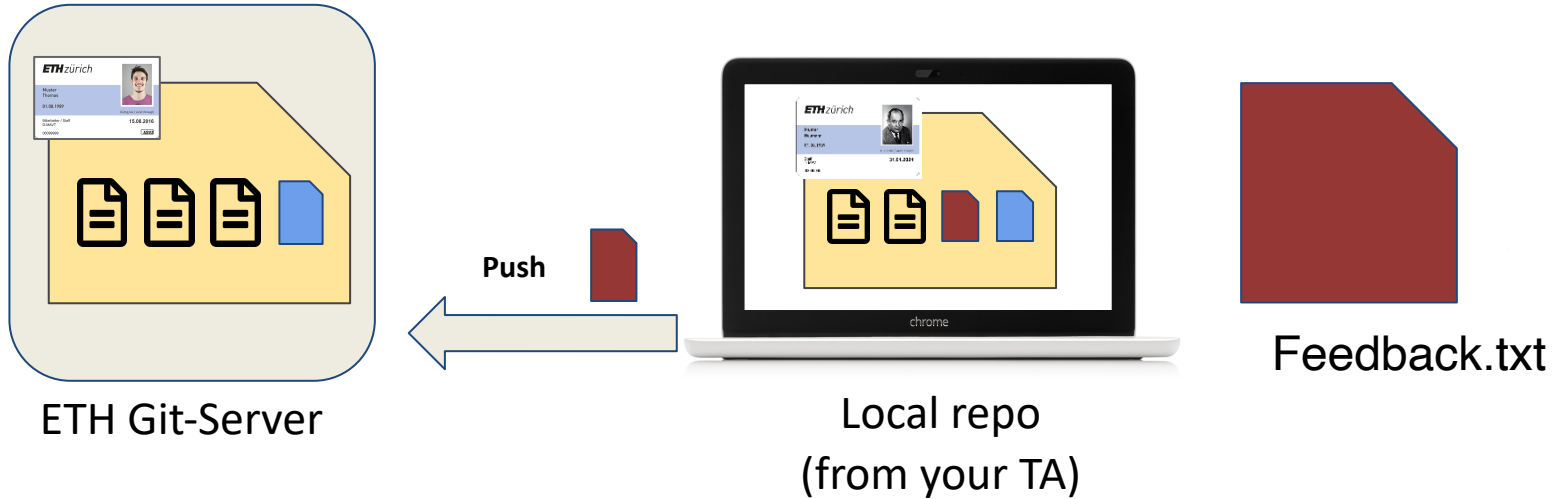
Git Pull: Load exercises / feedback



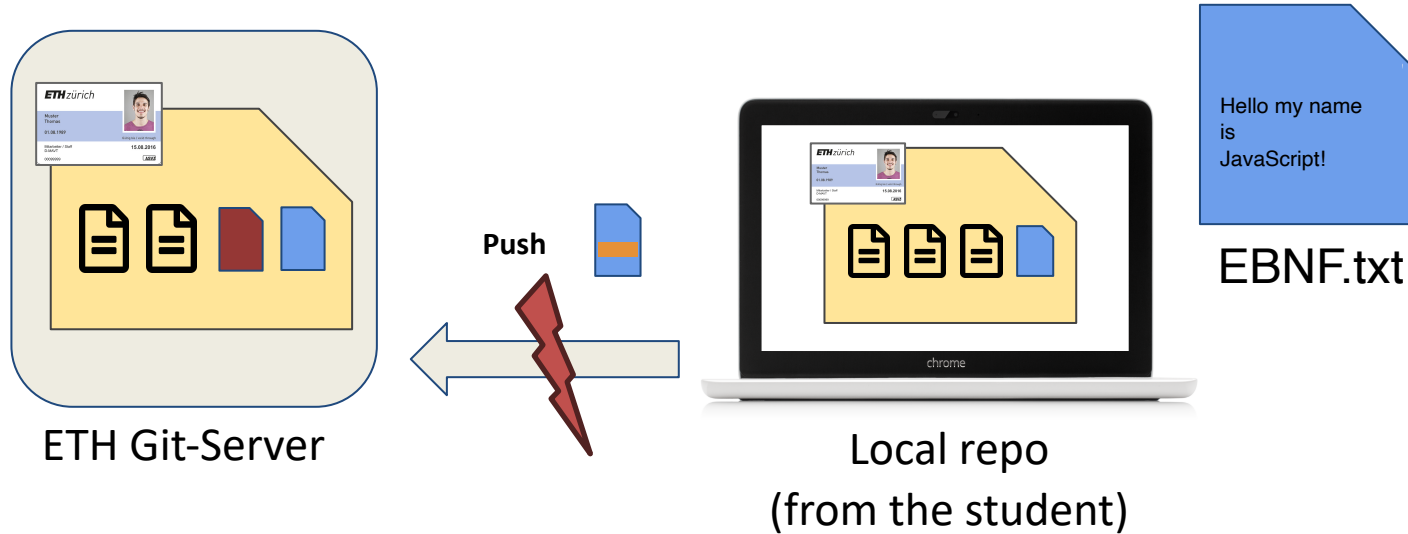
Git Pull/Push-Workflow



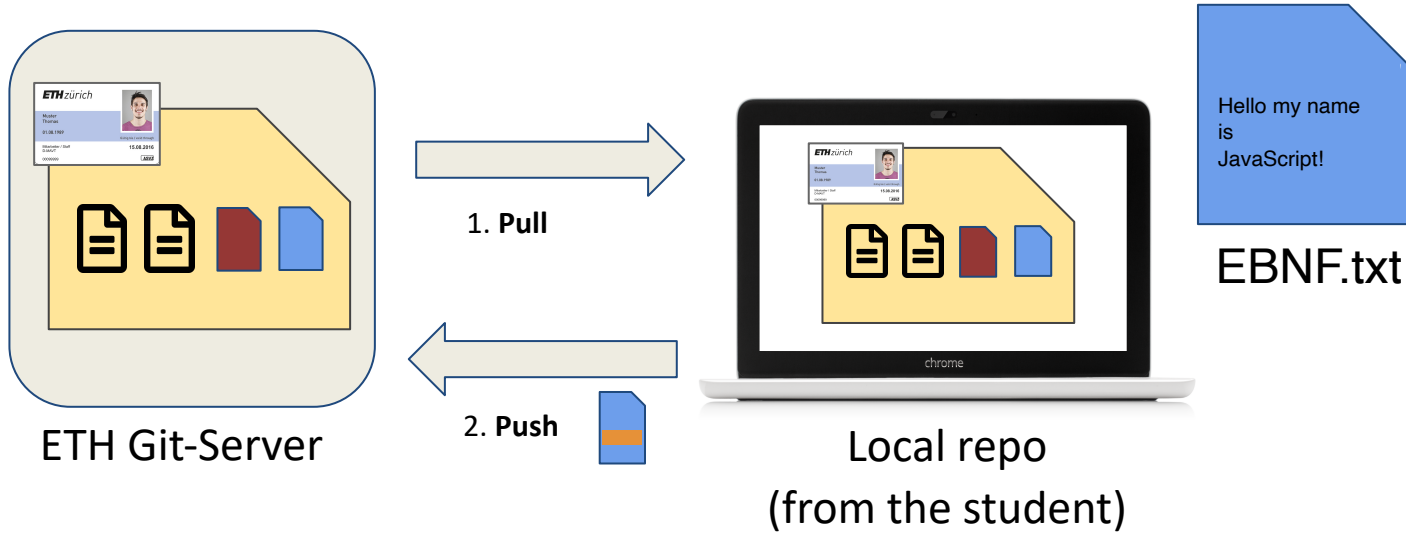
Git repository changes!



Git Merge Conflict



Prevent Git merge conflicts



Always pull first, then push!

Avoid common Git errors

- Log in to gitlab.inf.ethz.ch before first use
- In U02 Exercise 4, replace `<nethz-account>` in the link including `<>`
- In U02 Exercise 4, type the link by hand if there are problems
- Always delete or change files/directories/projects/... in IntelliJ (never in File Explorer/Finder)
- Always pull and then push (check online whether changes are visible)
- "Commit and Push" instead of just "Commit"
 - If you clicked Commit, right-click the repository manually and choose Git -> Push. Do not forget to import new projects!

Why EBNF

- Java Language Specification
- Chapter 19

EBNF Theory Recap

- Equivalence of EBNF Rules
 - String legal for B1 $\Leftrightarrow$ String legal for B2
 - String illegal for B1 $\Leftrightarrow$ String illegal for B2

EBNF Theory Recap

- Recursion
 - Direct Recursion
 - Name of the rule on RHS
 - Indirect Recursion
 - There is a List of rules such that name on the LHS is derivable on the RHS
- Recursion in EBNF is stronger than iteration

Extra Exercises

- Create a description `<palindrome>`, whose legal symbols allow all numbers that read the same from the front and the back and use only the digits from 1 to 4. Examples are 11, 232, 444

EBNF Notation

- In old exams, italics are often used for EBNF rules.
- *digit instead of <digit>*
- Starting this semester, an EBNF rule is only correct if it is marked by < >.
- We also use <- instead of <=, but both are graded as correct.

Kahoot!



forms.gle

forms.gle

Thank you very much for your attention

Have a great week!

EBNF: Legal / Not Legal

Given the EBNF description of value

digit $\Leftarrow$ 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9

separator $\Leftarrow$ -

buchst $\Leftarrow$ A | B | C | D | E | F | a | b | c | d | e | f

zahl $\Leftarrow$ *digit* { [*separator*] *digit* }

int $\Leftarrow$ *digit* { *digit* }

real $\Leftarrow$ *digit* { *digit* } [. *digit* { *digit* }]

bd $\Leftarrow$ *buchst* | *digit*

mix1 $\Leftarrow$ *bd* { *bd* }

mix2 $\Leftarrow$ *digit* { *digit* } k

mix $\Leftarrow$ *mix1* | *mix2*

value $\Leftarrow$ *zahl* | *real* | *int* | *mix*

2023

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Back

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1a2b3c

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12cd.34

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face

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0010K

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